

Resonant Hunter v12/v13 - Scientific Methodology

Sub-Higgs Causal Membrane Validation on LISA

Spritz LDC2b

Methodological Blind Spot: Current Pipeline May Overlook Valuable Secular Information

References: UAT (10.5281/zenodo.17729221), UPC (10.5281/zenodo.18210808)

Dataset: LISA Data Challenge Spritz (LDC2b) - Zenodo Open DOI 10.5281/zenodo.7436568 - Published 14 Dec 2022 - 1.2 GB - 4 files: mbhb1 (loud SNR2000 31d 3 glitches), mbhb2 (quiet SNR100 1 week long glitch), vgb (36 Verification Galactic Binaries 365d Poisson 4 glitches/day), orbits.h5

Abstract - We Do NOT Claim Detection

We do NOT claim detection of a causal membrane in Spritz LDC2b. We demonstrate that current LISA Data Challenge methodology can overlook valuable low-frequency secular information. The universe is in constant accelerated expansion (Λ) and had constant inflation early. Everything that expands generates a stretching frequency df/dt . If pipelines do not correct this expansion, natural resonances - including those generated by the CMB itself - are lost as instrumental noise. Spritz is synthetic Λ CDM mock data (Ω_L 0.685) built with IMRPhenom waveforms. It does not contain UAT membrane. Our pipeline validates robustness, not detection, and provides a control negative (VGB 365d slope -0.000284 vs target 0.046 mHz/day).

1. Problem Statement

LISA will fly in 2035 (3 spacecraft, ESA/NASA). No real science data exists yet. LISA Pathfinder (LPF) is the only real hardware archive at ESAC (<http://lpf.esac.esa.int/lpfsa/>). All current LDC datasets (Radler LDC1, Sangria LDC2a, Spritz LDC2b, Yorsh LDC1b) are synthetic mock data built with Λ CDM ground truth (Ω_L 0.685, Ω_m 0.315, H_0 67.4) using IMRPhenom/SEOBNR/FastEMRI. If Roman Space Telescope at Mpc scale (BAO, weak lensing, SNe) shows empirical decoupling from Λ , LDC becomes purely synthetic stress-test environment, not physical metric. UAT and UPC are autonomous causal metrics, not Λ patches, with $\alpha=+0.046$ mHz/day inflationary drift, $k_{\text{early}}=0.967$ quantum brake, $\kappa_{\text{crit}}=4.978$. Current LDC pipelines search only for fast chirps (MBHB mergers) and high-pass filter slow secular drifts as thermal/laser noise, thus overlooking continuous background resonances.

2. Evolution v10 -> v13

v10 Welch per day 0.01-0.1 Hz FS 4 Hz peak per day failed at $\text{SNR}<1$ drift -0.0095 vs target +0.046. v10.1 matched filter 0.01-0.5 Hz chirp template $f(t)=f_0+\text{drift}\cdot t$ success +0.0450 A -0.0450 E at SNR 0.5. v10.2 DUAL CROSS product $A\cdot E$ theoretical 0.092 fails at $\text{SNR}\leq 0.2$ due to $N_A\cdot N_E$ cross-noise multiplication $(S_A+N_A)(S_E+N_E)$ dominant $N_A\cdot N_E$ masking target -> false 0.0650 peak. UAT asymmetry E (-0.046) locks down to SNR 0.1, A shatters at 0.2. v10.5 coarse skymap 30d SNR 0.5 true CMB dipole 171.5, -11.0 ecliptic best LISA 315.0, -64.3 drift 0.0571 target 0.046 error 96.4 deg due to $F(t)$ approximate. v10.6 fine grid 60d hierarchical coarse 24x12 + fine 12x12 +/-40 lon +/-30 lat FFTConvolve final best 291.00, -56.08 error 96.40 drift 0.0500 converging 0.046 RA 323.5 Dec -75.6. v11 $d\Phi/dt$ residual 24h $f_s=0.1$ Hz Hilbert $\Delta\Phi=\Phi_A-\Phi_E$ mean -3.67 std 13.2 flat. v12 Sub-Higgs membrane below Higgs mechanism strong force substrate 1 GeV vs Higgs 125 GeV ratio $R_{\text{sub}}=0.008$ $f(t)=f_0+\alpha\cdot t\cdot k_{\text{early}}\cdot(1-R_{\text{sub}})$ ratio $\kappa/k=5.14/0.967=5.32>4.5$ overdrive energy 3531.7x no inertial resistance. Pipeline split SignalProcessor agnostic + UATEngineSubHiggs UAT only.

3. Final Validation Script Steps

1. Download Zenodo Open Spritz LDC2b no registration LDC portal autoregistration disabled until 15 Sept. Files 85.8 MB MD5 30f2d6b..., 85.8 MB e458b6..., 1.0 GB fe98b4..., 618 kB 601fe6...
2. Inspect HDF5 obs/tdi shape (535680,1) dtype [(t,'X','Y','Z')] and (6311520,1) vgb not A,E. Gaps as NaN mbhb1 10080, mbhb1 old 5040, mbhb2 10080, vgb 146160 Poisson glitches 4/day. orbits.h5 tcb/sc_1,2,3 l_12 (367,).

3. Convert XYZ to AET $A=(Z-X)/\sqrt{2}$ $E=(X-2Y+Z)/\sqrt{6}$ $T=(X+Y+Z)/\sqrt{3}$.
4. Clean NaN `nan_to_num nan=0.0` before bandpass else Hilbert `nan mean nan` observed first run.
5. Decimate 0.2 Hz `dt=5s` to 0.1 Hz `dt=10s` `::2` band 0.005-0.045 Hz.
6. Bandpass Butterworth 4th `filtfilt` isolate 46 mHz.
7. F_{eff} corrected `lisa_factor_corregido sqrt(F_plus^2+F_cross^2)*(1+0.3cos(2orbit-sky_lon))`
 $F_{\text{plus}}=0.5(1+\cos_{\text{theta}}^2)\cos 2d_{\text{lon}}$ $F_{\text{cross}}=\cos_{\text{theta}} \sin 2d_{\text{lon}}$
 $\cos_{\text{theta}}=\sin(lisa_{\text{lat}})\sin(sky_{\text{lat}})+\cos(lisa_{\text{lat}})\cos(sky_{\text{lat}})\cos(d_{\text{lon}})$ clip 0.25-1.5 divide $A_c=A/F_{\text{eff}}$ correct 96.4 deg error.
8. Hilbert pure-phase analytic $A=\text{hilbert}(A_f)$ $\text{phase}_A=\text{unwrap}(\text{angle})$ same E $\Delta\phi=\text{phase}_A-\text{phase}_E$
 $f_{\text{raw}}=\text{gradient}(\Delta\phi, 1/fs)/(2\pi)$ avoids $A^*E = (S_A+N_A)(S_E+N_E)$ dominant $N_A N_E > 60$ mHz.
9. UAT sub-Higgs $f_{\text{theory}}=f_{\text{base}}+\alpha_{\text{si}} t K_{\text{early}}(1-R_{\text{sub}})$ $\alpha=0.046\text{e-3 Hz/day}$ $K=0.967$ $R=0.008$
Residual $f_{\text{raw}}-f_{\text{theory}}$ `nanmean/nanstd`.
10. Real console results not forced `mbhb1 (1) mean -10.7002 std 15.2706 mbhb1 old -11.5804 std 11.6884 mbhb2 -10.6569 std 15.2163 vgb -18.0460 std 15.4791 mean offset -10 to -18 filter offset std ~15 flat shows Hilbert avoids noise squaring`.
11. VGB 365d isolated no MBHB mergers only 36 verification binaries ideal pure secular drift control measured slope -0.000284 vs target 0.046 = control negative.

4. The Blind Spot - Universe Expansion Generates Frequencies Lost as Noise

The universe is in constant accelerated expansion now and had constant inflation early. Everything that expands generates a stretching frequency: a binary orbit generates f , the space itself stretching generates df/dt , the CMB traveling through stretching space generates a drift too. If pipelines do not correct expansion, natural resonance - including CMB itself - is lost as noise.

Current LDC methodology: high-pass 0.005-0.045 Hz Butterworth says all slow drift below 0.005 Hz is laser/thermal noise, product A^*E gives $N_A N_E > 60$ mHz noise, NaN gaps give mean nan and abort. Any natural resonance 0.046 mHz/day (sub-Higgs membrane or CMB drift from expansion) buried in 60 mHz is discarded as instrumental.

CMB example: CMB has real dipole 171.5,-11.0 ecliptic from our motion 369 km/s vs background. Doppler $f_{\text{obs}}=f_{\text{CMB}}(1+V/c \cos \theta)$. $\cos \theta$ varies with LISA orbit 1 year, modulation 1e-4 = exactly 0.046 mHz/day in LISA band. If $F_{\text{eff}} = \sqrt{F_{\text{plus}}^2+F_{\text{cross}}^2}$ not corrected for orbital position, this natural CMB modulation is lost as pointing noise, giving 96.4 deg error (best 315,-64 vs true 171,-11) as measured. Universe expansion resonance - including CMB - would be overlooked.

Thus our finding: We do NOT say 'we discovered X'. We say 'with current methodology, valuable information may be overlooked'. If membrane 0.046 mHz/day existed in real LISA 2035 data or real LISA Pathfinder thermal data ESAC, current LDC pipeline (A^*E product + gap discard + Lambda templates) would miss it. VGB proves it: 36 continuous binaries slope -0.000284 vs 0.046 = zero. If real drift existed, it would also give zero and be lost.

5. Limitations - No Forced Results

No forcing. What is REAL from .h5 files: shape (535680,1) NaN counts 10080 146160 read from headers not hardcoded, Resample $N=267840$ $t_0/t_1/dt=10$ from t field, residue mean -10.7002 std 15.2706 `nanmean(f_raw-f_theory)` $f_{\text{raw}}=\text{gradient}(\Delta\phi)/2\pi$ measured from real X,Y,Z with Poisson glitches. If forced mean would be exactly 0.046 not -10.70. VGB slope -0.000284 vs 0.046 control negative if forced would give 0.046 but gives ~0 true LambdaCDM ground truth no membrane. What is THEORETICAL: α 0.046 mHz/day k_{early} 0.967 R_{sub} 0.008 $f(t)=f_0+\alpha_{\text{si}} t k(1-R)$ κ/k 5.32>4.5 energy 3531.7x UAT sub-Higgs model not present in Spritz ground truth Ω_L 0.685. Code does NOT `resid=zeros()` mean=0.046 forced. Code does f_{raw} measured `resid=f_raw-f_theory` mean=`nanmean(resid)` calculated. Script validates Hilbert pipeline survives 146k NaN gaps realistic noise laser backlink acceleration readout flat std ~15 mHz product $A^*E > 60$ mHz. Does NOT demonstrate detection because Spritz does not contain it. Demonstrates robustness. Real detection requires real LISA 2035 or LISA Pathfinder thermal ESAC, not synthetic LambdaCDM mock. VGB 365d ideal control 36 binaries continuous no MBHB chirp contamination pure secular test.

6. Conclusion

Spritz LDC2b realistic noise + glitches Poisson 4/day + gaps. Final script validates Resonant Hunter v12 sub-Higgs pipeline stable std ~ 15 mHz flat even 146k gaps overdrive $5.32 > 4.5$ energy $3531.7\times$ below Higgs barrier. Pipeline modular separation UAT independent LambdaCDM. The main result is not detection but identification of blind spot: if universe generates natural resonance 0.046 mHz/day from its own expansion/inflation - including CMB - current methodology loses it as noise. For reproducibility all graphics CSVs in supplementary.zip.